Using Methods

Writing your own methods

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Topics list

1. Quick Recap of method **terminology**:
   – Return type
   – Method names
   – Parameter list

2. **Writing your own** methods:
   – With no parameters
   – With parameters
   – That return data

Recap: Methods in Processing

- A method is comprised of
  a **set of instructions that performs some task**.

- When we invoke **(call)** the method, it performs the task.

- Some methods we have used are:

  - **main** – (we write the code for main)
  - **print(), println() –** we call these methods

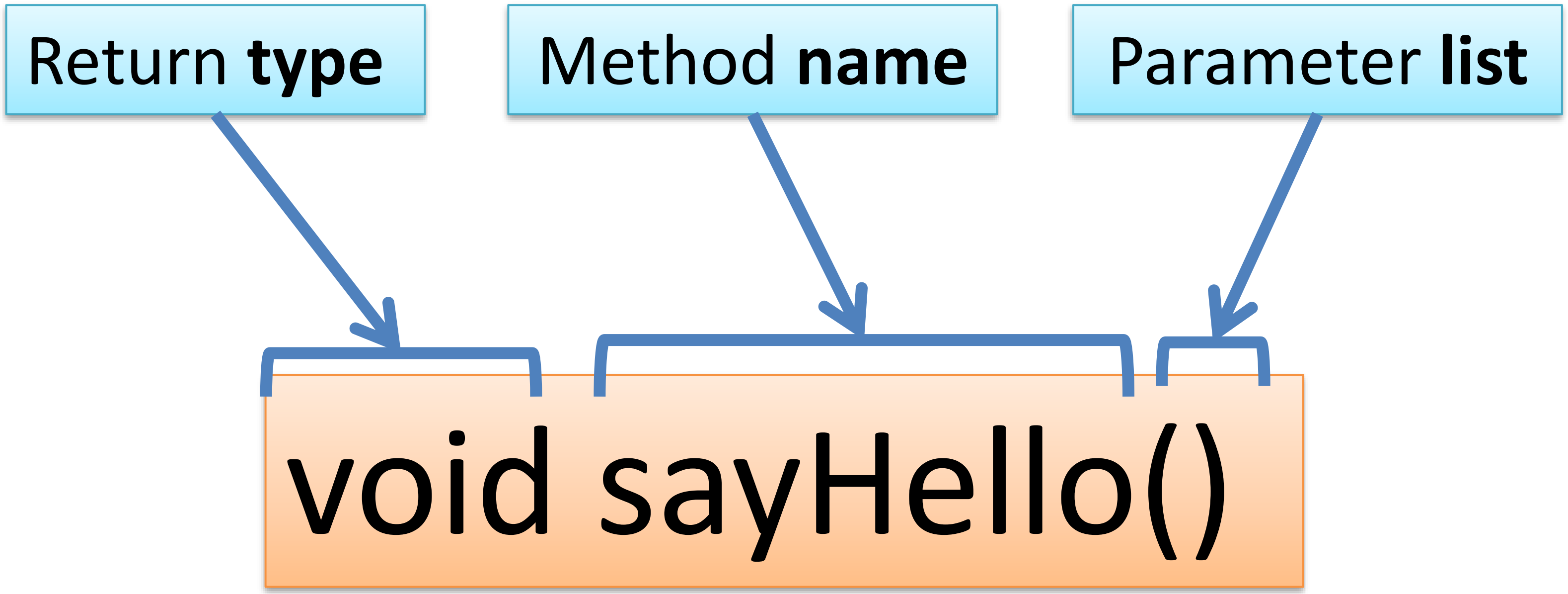
Recap: Method terminology

Method **signature / header**

Method **body**

```
void sayHello()
{
   System.out.println(“hello”);
}
```

Recap: Method **signature**



Topics list

1. **Recap** of method **terminology**:
   – Return type
   – Method names
   – Parameter list

2. **Writing your own** methods:
   I. With no parameters
   II. With parameters
   III. That return data

Writing methods **with NO parameters**

```
Hello!
```

- Write a method that prints a hello message,
  - every time it is called (invoked) it will print the message.

Example 3.1

Hello!

Method definition

```java
import java.util.Scanner;

public class MethodExamples {  2 usages
    // 1 Method with NO parameters and NO return
    public  void sayHello() {  1 usage
        System.out.println("Hello!");
    }

}
```

Create an instance of the class

Method call

```java
public class Driver {
    // Main method to test them all
    public static void main(String[] args) {
        MethodExamples me = new MethodExamples(); //create an object
        // 1 Call method with no parameters / no return
        me.sayHello();
    }

}
```

Topics list

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2. **Writing your own** methods:
   1. With no parameters
   2. With parameters
   3. That return data

Writing methods **with parameters**

```
Hello!
Hello, Mairead!
```

- Now update the code so that you can:

  pass in a String to hold a name – this method will print a message using that name

Example 3.2

```
Hello!
Hello, Mairead!
```

MethodExamples.java

```java
import java.util.Scanner;

public class MethodExamples {  2 usages
    // 1 Method with NO parameters and NO return
    public  void sayHello() {  1 usage
        System.out.println("Hello!");
    }
    // 2 Method with PARAMETERS and NO return
    public  void greet(String name, int age) {  1 usage
        System.out.println("Hello, " + name + ", you are "
                + age + " years old.");
    }

}
```

Driver.java

```java
public class Driver {
    // Main method to test them all
    public static void main(String[] args) {
        MethodExamples me = new MethodExamples(); //create an object
        // 1 Call method with no parameters / no return
        me.sayHello();
        // 2 Call method with parameter / no return
        me.greet( name: "Mairead",  age: 21);
    }
}
```

Using Scanner **Example**

```
Hello!
Hello, Mairead!
Enter a name:
Siobhan
Hello, Siobhan!
```

MethodExamples.java

```java
import java.util.Scanner;

public class MethodExamples {  2 usages
    // 1 Method with NO parameters and NO return
    public  void sayHello() {  1 usage
        System.out.println("Hello!");
    }
    // 2 Method with PARAMETERS and NO return
    public  void greet(String name, int age) {  1 usage
        System.out.println("Hello, " + name + ", you are "
                + age + " years old.");
    }

}
```

```java
// Main method to test them all
public static void main(String[] args) {
    MethodExamples me = new MethodExamples();
    Scanner sc = new Scanner(System.in);
    // 1 Call method with no parameters / no return
    me.sayHello();
    me.greet( name: "Mairead");
    System.out.println("Enter a name: ");
    String name = sc.nextLine();
    me.greet(name);
}
```

Writing methods **with parameters**

```
Hello!
Hello, Mairead, you are 21 years old.
```

- Now update the code so that you can pass in the:
  - **name**
  - **age**

into the method, *greet().*

Example 3.3

```
Hello!
Hello, Mairead, you are 21 years old.
```

```java
public class MethodExamples {
    // 1 Method with NO parameters and NO return
    public  void sayHello() {  1 usage
        System.out.println("Hello!");
    }
    // 2 Method with PARAMETERS and NO return
    public  void greet(String name, int age) {  1 usage
        System.out.println("Hello, " + name + ", you are " + age + " years old.");
    }
    // Main method to test them all
    public static void main(String[] args) {
        MethodExamples me = new MethodExamples();
        Scanner sc = new Scanner(System.in);
        // 1 Call method with no parameters / no return
        me.sayHello();
        me.greet( name: "Mairead",  age: 21);
    }
}
```

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Writing <u>methods **that return data**</u>

- Write a method called **timesTwo.**

- This method should
  - **take in one int parameter**.
  - **multiply this int by 2** and
  - **return** it back to where the **timesTwo** method was called from.
  - The returned value should be **printed to the console**.

Example 3.4

Alternative implementations of **timesTwo**:

```
int timesTwo(int val){
   val = 2 * val;
   return val;
}
```

```
int timesTwo(int val){
   val *= 2;
   return val;
}
```

```
int timesTwo(int val){
   return val *= 2;
}
```

```
public class MethodExamples {  2 usages
    // 1 Method with NO parameters and NO return
    public  void sayHello() {  1 usage
        System.out.println("Hello!");
    }
    // 2 Method with PARAMETERS and NO return
    public  void greet(String name, int age) {  1 usage
        System.out.println("Hello, " + name + ", you are "
                + age + " years old.");
    }
    // 3 Method with PARAMETERS and WITH return
    public  int timesTwo(int a) {  1 usage
        return a * 2;
    }
}
```

```
public class Driver {
    // Main method to test them all
    public static void main(String[] args) {
        MethodExamples me = new MethodExamples(); //create an object
        // 1 Call method with no parameters / no return
        me.sayHello();
       // 2 Call method with parameter / no return
        me.greet( name: "Mairead",  age: 21);
        // 3 Call method with parameters / with return
        int sum = me.timesTwo( a: 5);
        System.out.println("5 * 2 = " + sum);
    }
}
```

Summary

1. Recap of method **terminology**:
   – Return type
   – Method names
   – Parameter list

2. **Writing your own** methods:
   – With no parameters
   – With parameters
   – That return data

Questions?